Multi-Object Detection and Persistent ID Tracking

1. Introduction

This project aims to build a computer vision pipeline capable of detecting multiple objects and assigning persistent IDs across video frames. The system is designed to handle real-world challenges such as occlusion, motion, and camera movement using modern detection and tracking techniques.

2. Model Used (Object Detection)

The object detection task is performed using **YOLOv8 (You Only Look Once)** from the Ultralytics library.

- It is a state-of-the-art real-time object detection model.
- It provides high speed and good accuracy.
- Pre-trained weights were used for detecting objects in video frames.

YOLOv8 detects multiple classes, but in this project, only the **person class** is considered to focus on tracking players.

3. Tracking Algorithm Used

For multi-object tracking, the **ByteTrack algorithm** (via the Supervision library) is used.

- ByteTrack associates detections across frames.
- It assigns unique IDs to each object.
- It handles temporary object disappearance and reappearance.

4. Approach / Pipeline

The complete pipeline consists of the following steps:

1. Input video is read frame by frame
2. YOLOv8 detects objects in each frame
3. Only person detections are filtered

4. Low-confidence detections are removed
5. Very small bounding boxes are filtered out
6. ByteTrack assigns and maintains object IDs
7. Trajectories are stored and drawn
8. Player count is calculated per frame
9. Final annotated video is generated

5. ID Consistency Strategy

To improve tracking stability and reduce ID switching:

- Only person class detections are used
- Confidence threshold filtering is applied
- Small bounding boxes are removed
- Trajectory history is maintained
- ByteTrack handles ID association across frames

Despite these improvements, ID switches may still occur due to complex real-world conditions.

6. Challenges Faced

- Occlusion between players
- Similar appearance of players
- Fast movement in sports scenes
- Camera motion and zoom
- ID switching during overlaps

7. Failure Cases

- ID switching when players overlap heavily

- Temporary loss of tracking when players exit frame
- Confusion in crowded scenes

8. Possible Improvements

- Use DeepSORT with Re-Identification (ReID)
- Train a custom detection model
- Use higher resolution video
- Add team classification
- Apply motion prediction models

9. Conclusion

This project successfully demonstrates a real-time multi-object detection and tracking system. The integration of YOLOv8 and ByteTrack provides an efficient solution for tracking multiple subjects with reasonable accuracy.

10. Video Source

(https://youtu.be/6CS_mU8lj3I?si=3ZI82MbBM3trHem6)